Genre Classification of Spotify Tracks

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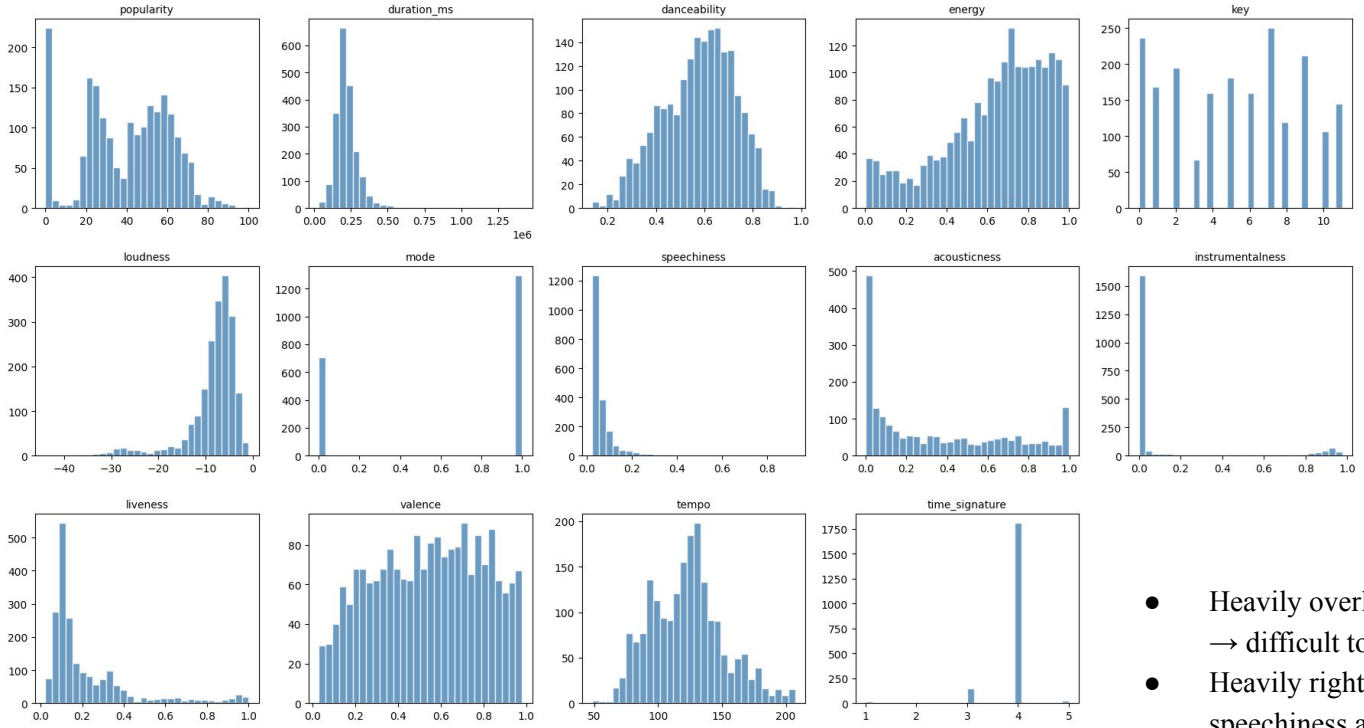
Introduction

- Dataset from Kaggle included 114,000 Spotify tracks and 21 features.
- Selected the top 10 most common genres and sampled 200 songs per genre
- 3 Models: Logistic Regression, SVM, Neural Network

Song Genre

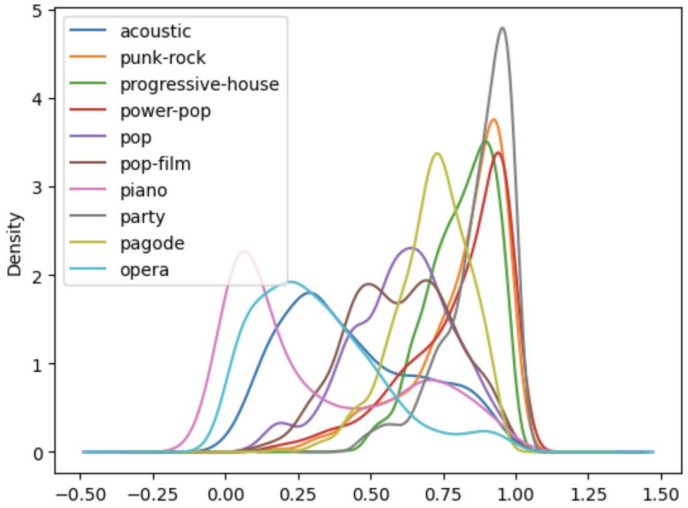
Your Favorite Song Genre

Challenge: Many genres share similar musical features, causing overlap between classes



Feature Distribution of the Dataset

- Heavily overlapping ranges (danceability and tempo) → difficult to use them as reliable discriminators.
- Heavily right-skewed distributions, such as speechiness and acousticness →motivated log transformation → helps spread out the dense region near zero and gives the model a more balanced view of those features
- Mode and time_signature show near-constant values → carry very little help for genre classification



Kernel Density Estimate for ‘Energy’

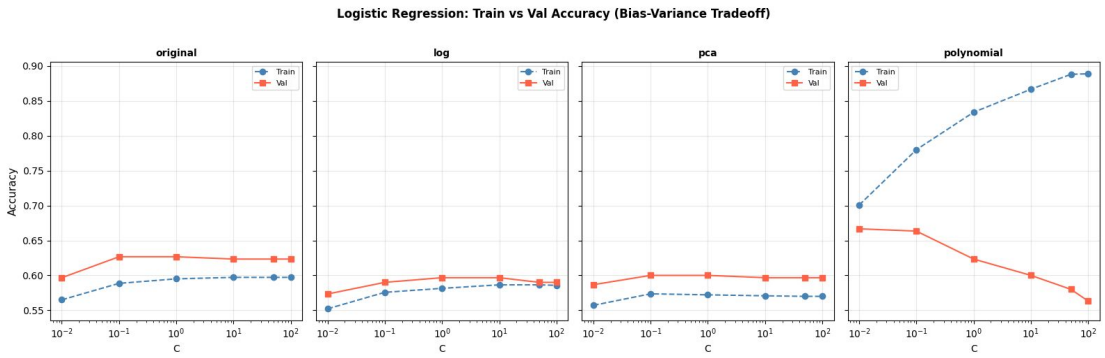
- Show how much they blur together
- Curves are overlapping proves that one feature does not cleanly separate genre
- A peak at 1 for some genres, showing some common trends
- Shows the biggest ‘issue’ and challenge with the data

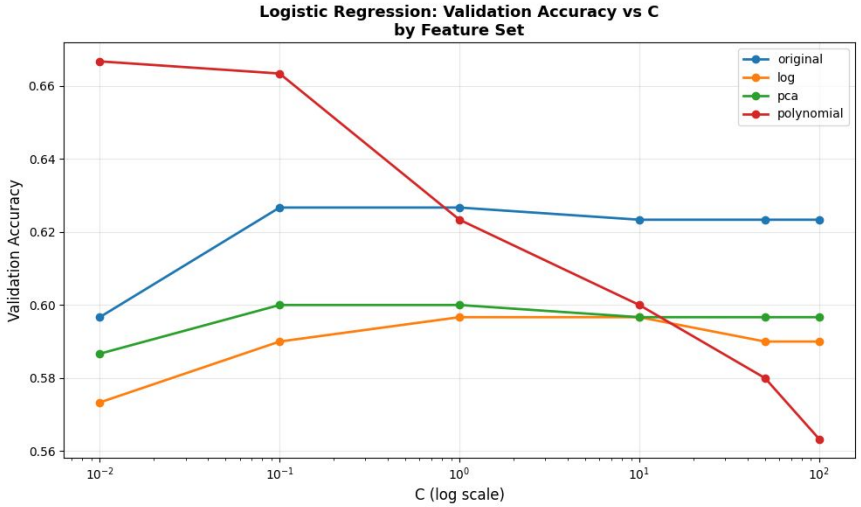
Preprocessing

- Data quality was strong: there were almost no missing values.
- Created a balanced dataset of 2,000 songs
- Used key numeric audio features for modeling, including popularity, duration, danceability - 14 features
- Converted genre labels from text into numbers using LabelEncoder
- 70% training / 15% validation / 15% test ratio
- Applied stratified sampling
- Standardized features using StandardScaler

Logistic Regression

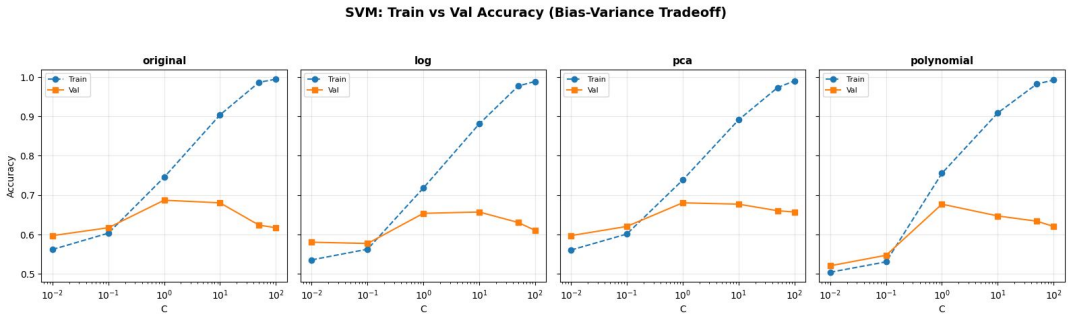
- Feature sets:
  - Original scaled features
  - PCA-transformed features
  - Polynomial features, degree = 2
  - Log-transformed features
- L2 Regularization
  - C = [0.01, 0.1, 1, 10, 50, 100]
- Best: polynomial transformation with C = 0.01
  - 66.7% validation accuracy
  - 63% test accuracy
  - 66.8 F1 score

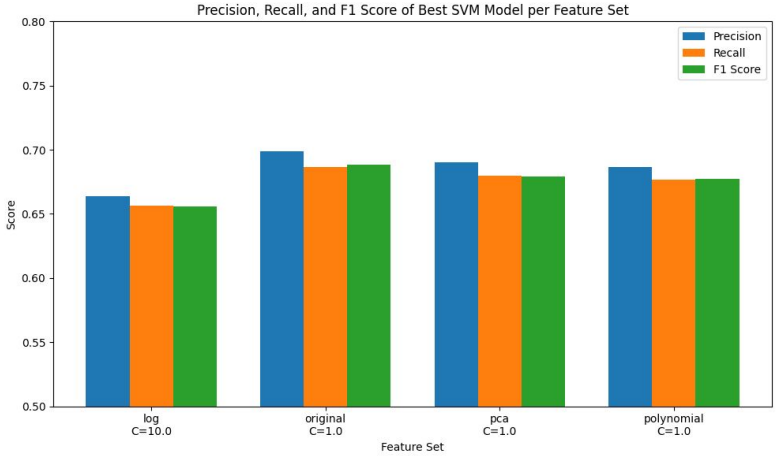




Support Vector Machine (SVM)

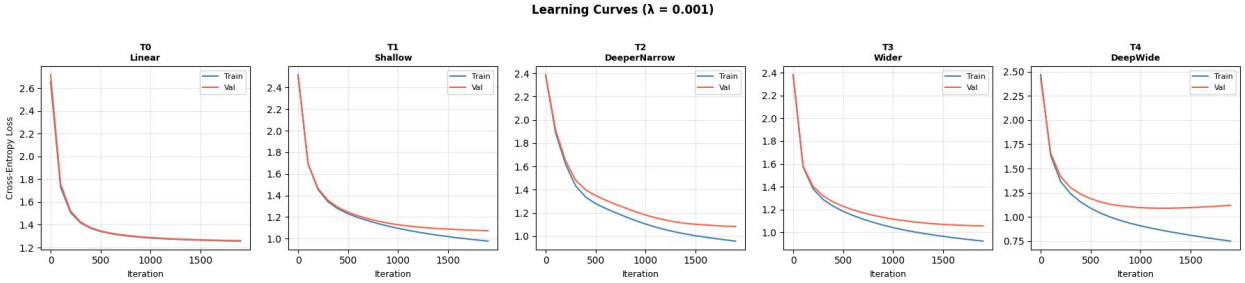
- Feature sets:
  - Original scaled features
  - PCA-transformed features
  - Polynomial features, degree = 2
  - Log-transformed features
- L2 Regularization
  - C = [0.01, 0.1, 1, 10, 50, 100]
- Best: PCA feature set with C = 1
  - 66% test accuracy
  - 68% validation accuracy
  - Original features with C = 1
    - 65.6667% test accuracy
    - 68.6667% validation accuracy



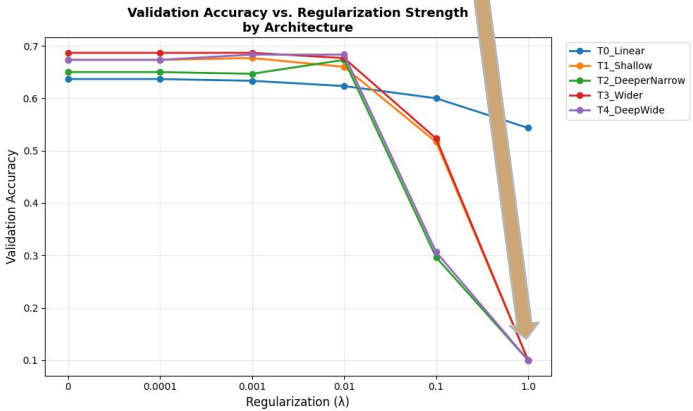


Neural Network

- Feature sets:
  - Linear: [14, 10]
  - Shallow: [14, 30, 10]
  - DeeperNarrow: [14, 20, 20, 10]
  - Wider: [14, 64, 10]
  - DeepWide: [14, 64, 32, 10]
- L2 Regularization
  - λ = [0, 0.0001, 0.001, 0.01, 0.1, 1]
- Best: Wider architecture with λ = 0
  - 68.67% validation accuracy
  - 67.7% test accuracy
  - 68.2% test recall
  - 67.4% test F1 score



When λ = 1.0, the regularization was so strong that most of the neural networks were forced into almost the same simple behavior, no matter how deep or wide they were



Conclusion

- Logistic Regression
  - Polynomial features gave the strongest performance
  - Original, log, and PCA logistic models stayed relatively similar
  - Feature transformation mattered more than regularization
  - Weakly regularized polynomial logistic regression quickly overfit
- SVM
  - PCA gave the strongest SVM test result
  - Log transformation removed useful SVM signal
  - SVM performed best with moderate regularization
  - SVM worked well without heavy feature transformation
  - Larger C → overfitting; Smaller C → underfitting
- Neural Network
  - Wider neural network performed best
  - DeepWide fit training best, but overfit more clearly
  - Bigger networks helped, but gains quickly leveled off
  - Strong regularization caused underfitting in neural networks, and vice versa

62-65%

Accuracy ceiling

Limitations

- Data limits and genre overlap capped performance overall
- Logistic regression struggled with a high feature-to-sample ratio
- Neural networks showed the strongest effects of data scarcity
- SVM needed more data for clearer decision boundaries

Improvements

- L1 regularization could select useful polynomial features
- Degree-3 features may reveal richer genre interactions
- Tuning SVM gamma could improve kernel performance
- Dropout or early stopping could reduce neural network overfitting
- Running neural networks multiple times with different initial weights would improve result reliability

Work Cited

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